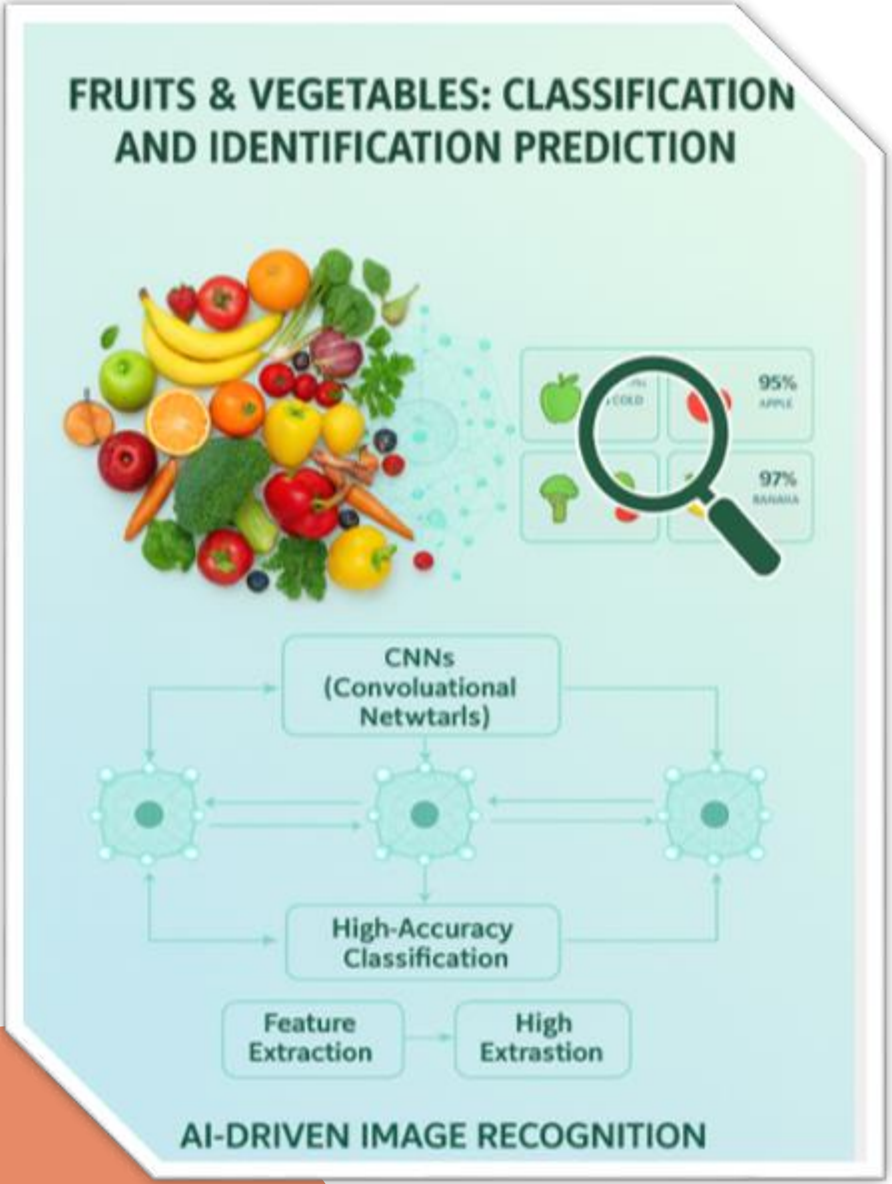
Project Title

Deep Fruit Veg Classification & Identification



Introduction :

Deep Fruit Veg is an AI-driven image recognition system developed using **Machine Learning** and **Deep Learning** techniques. It employs **Convolutional Neural Networks (CNNs)** to extract and analyze visual features from fruit and vegetable images, enabling high-accuracy classification for both single and multiple inputs. The model is integrated into a **Flask-based web application**, ensuring easy deployment, fast processing, and a user-friendly interface suitable for research, industry, and educational use cases.

Objective

- **Build an intelligent AI model for fruit and vegetable recognition**

  Develop a high-accuracy deep learning model capable of identifying multiple fruit and vegetable types reliable.

- **Replace slow, manual classification with automation.**

  Speed up sorting and recognition processes while reducing human effort and human error.

- **Support both single and multiple image processing**

  Provide flexibility to classify individual items or batches of images with consistent accuracy.

- **Deliver a clean and user-friendly web application**

  Offer a simple, responsive Flask-based interface suitable for students, retailers, farmers, and researchers.

- **Enhance efficiency, consistency, and quality control**

  Ensure precise, repeatable results that improve workflows in supermarkets, agriculture, and food-sorting systems.

- **Enable future scalability and improvements**

  Create a modular system capable of expanding to more categories, more datasets, and advanced models.

Problem Statement :

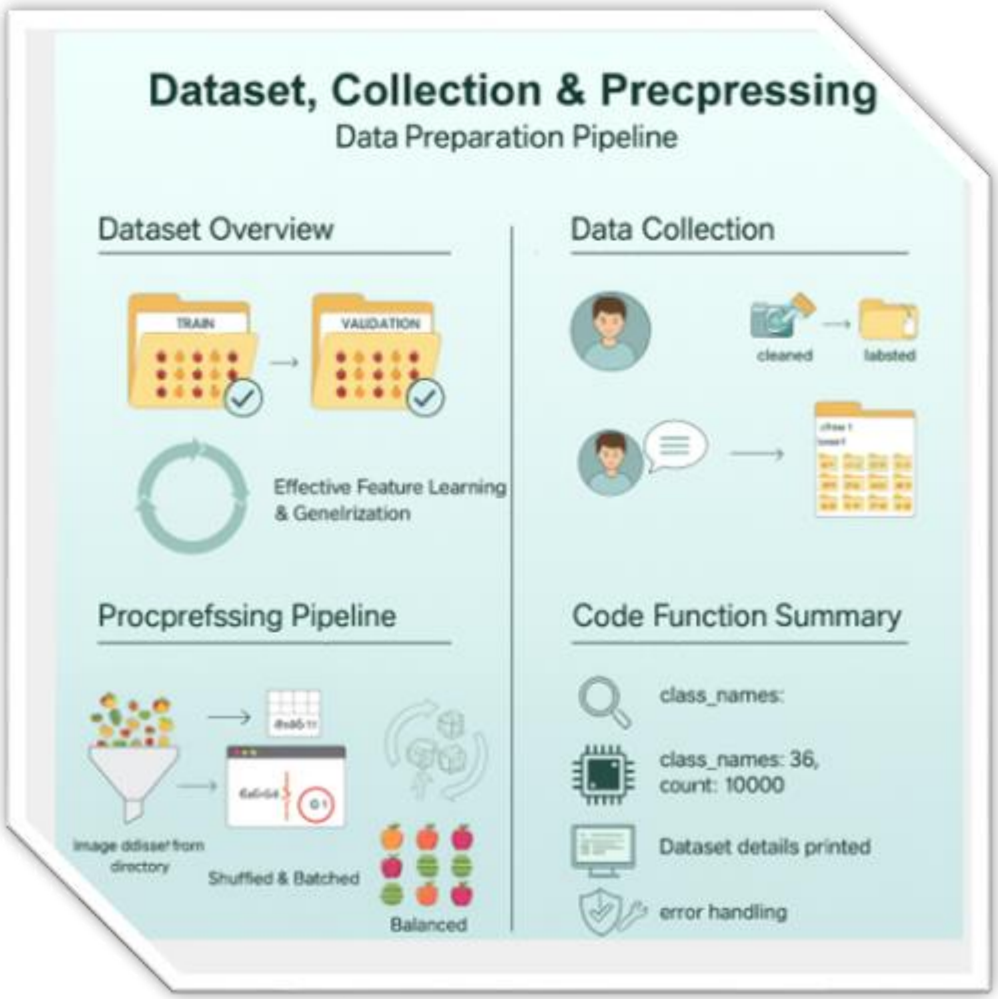
- **Challenges in Identifying Fruits & Vegetables**
  - Manual classification is time-consuming
  - Similar-looking items cause confusion
  - Need for fast, automated identification
  - Important for supermarkets, agriculture & food sorting

- **Why This Project? (DeepFruitVeg Solution)**

  DeepFruitVeg is built to solve these challenges by offering:

  - Automated, fast, and accurate AI-based classification
  - Reliable detection even for similar-looking items
  - Reduced human effort and improved workflow efficiency
  - A simple, user-friendly Flask web interface for real-world use

Dataset, Collection & Preprocessing



➢ **Dataset Overview**

Contains multiple fruit & vegetable classes, Split into **Train** and **Validation** folders, Supports effective feature learning & generalization.

➢ **Data Collection**

Images collected, cleaned, and labeled .Proper folder structure ensures correct class mapping.

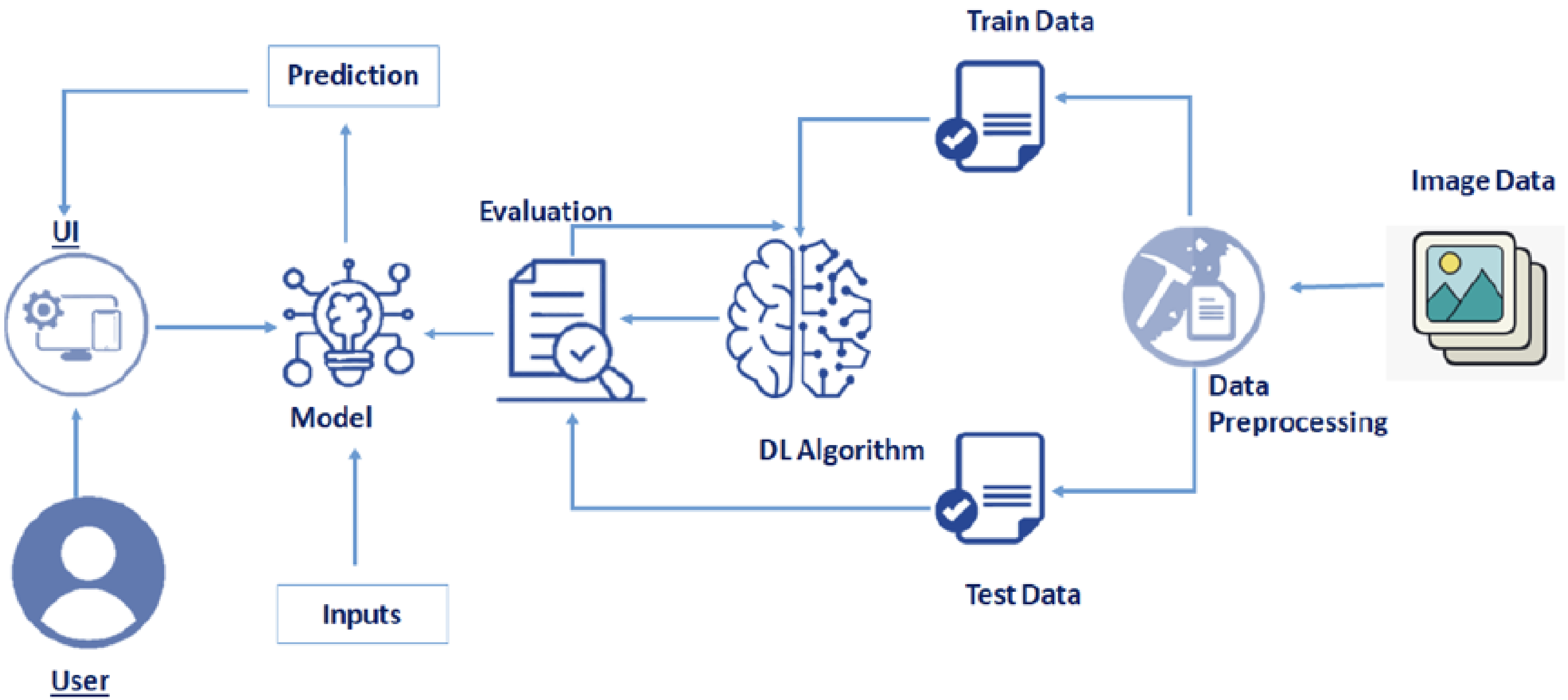
➢ **Preprocessing Pipeline**

Loaded using **image_dataset_from_directory,** Images resized to **64×64** and normalized, Shuffled & batched for stable learning, Dataset balanced for equal class representation.

➢ **Code Function Summary**

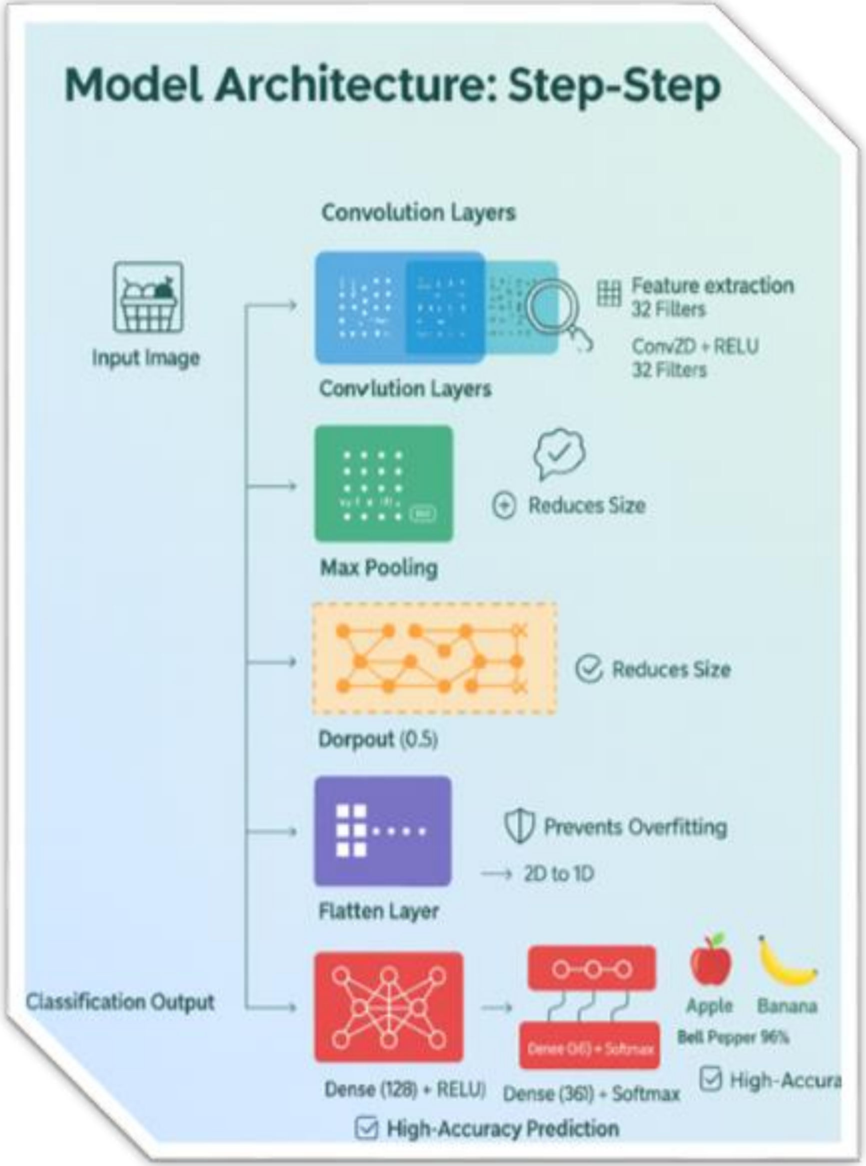
Auto-detects class names and counts, Converts images to tensors for training, Prints dataset details for verification, Built-in error handling ensures smooth loading.

Model Architecture :



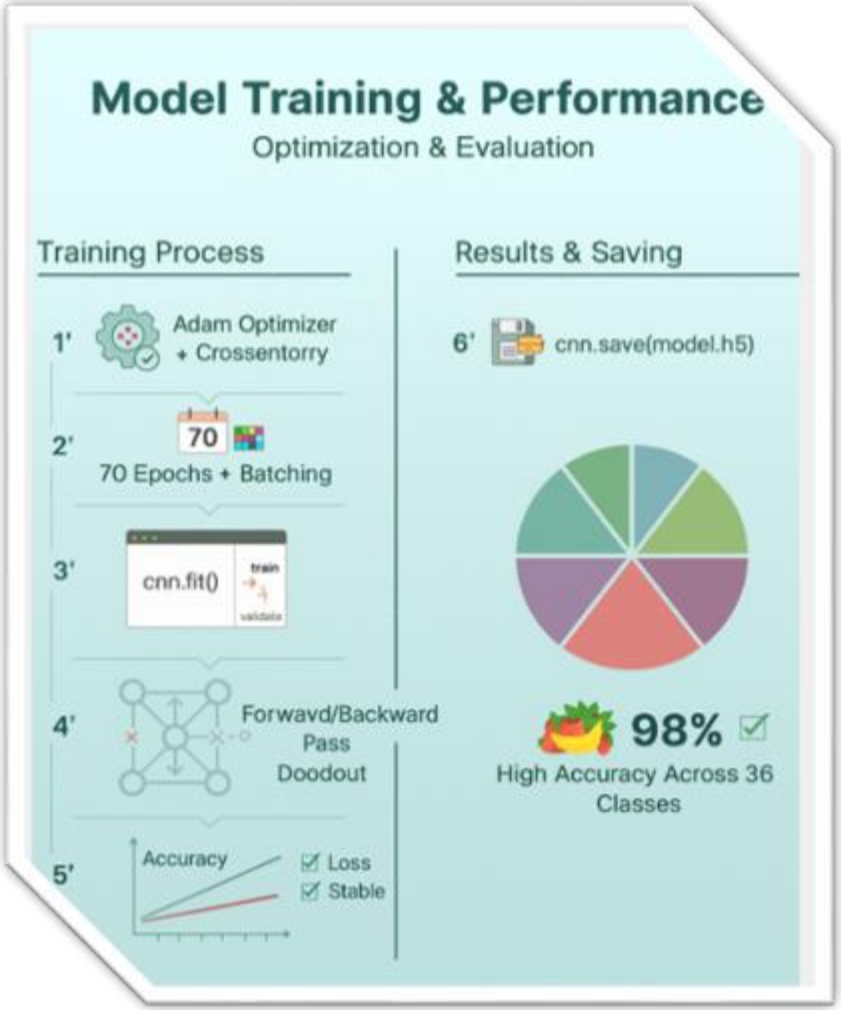
Model Building :

- **Model Overview** :
- Deep Learning model built using Convolutional Neural Network (CNN) Designed for accurate fruit &vegetable multi-class classification Uses feature extraction + dense classification pipeline
- 1 **Convolution Layers :**
- Two Conv2D layers (32 filters, 3×3, relu) extract key **texture, color, and shape features**, preserving spatial detail.
- 2 **Max Pooling Layer :**
- A 2×2 Max Pool layer **reduces spatial size and computation**, keeping only the most important visual information.
- 3 **Dropout Layer (0.5) :**
- A 50% dropout rate **prevents overfitting** by randomly disabling neurons for better generalization.
- 4 **Flatten Layer :**
- Flattens feature maps into a **1D vector**, enabling transition to dense classification layers.
- 5 **Dense Layers :**
- A 128-unit relu Dense layer captures deeper patterns, followed by a **36-class Soft max output** for final prediction.



Model Training, Optimization & Performance :

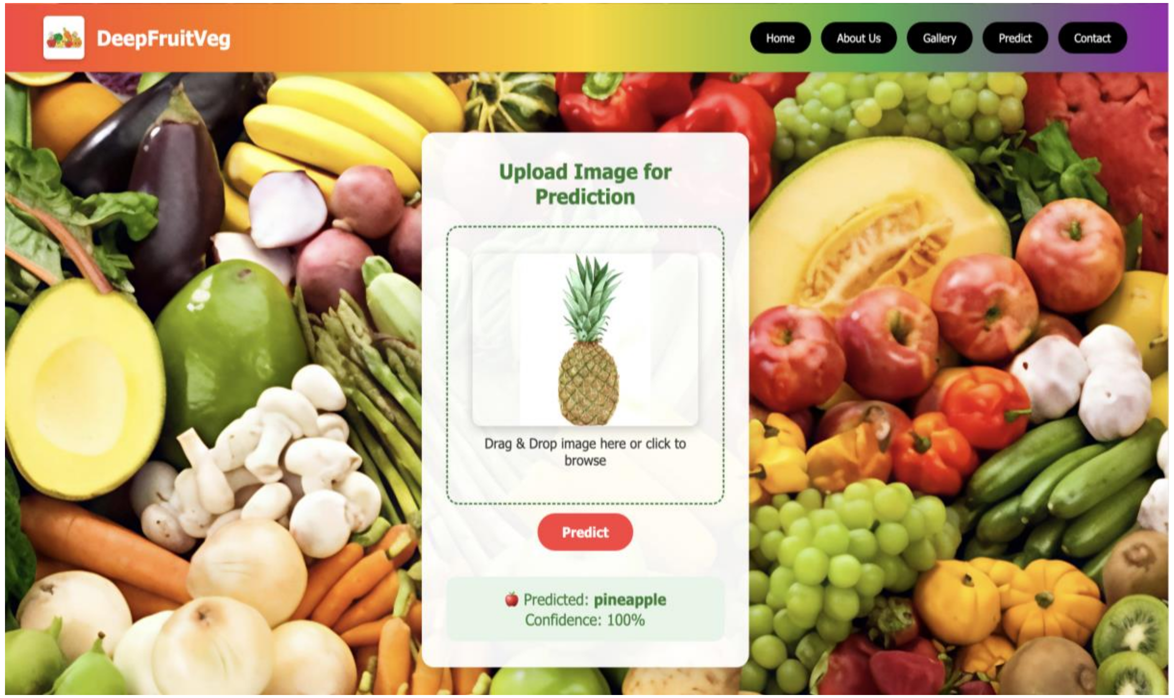
Model Compilation
The model is compiled using the **Adam optimizer** along with **categorical crossentropy loss**, ensuring efficient multi-class learning with adaptive weight updates.



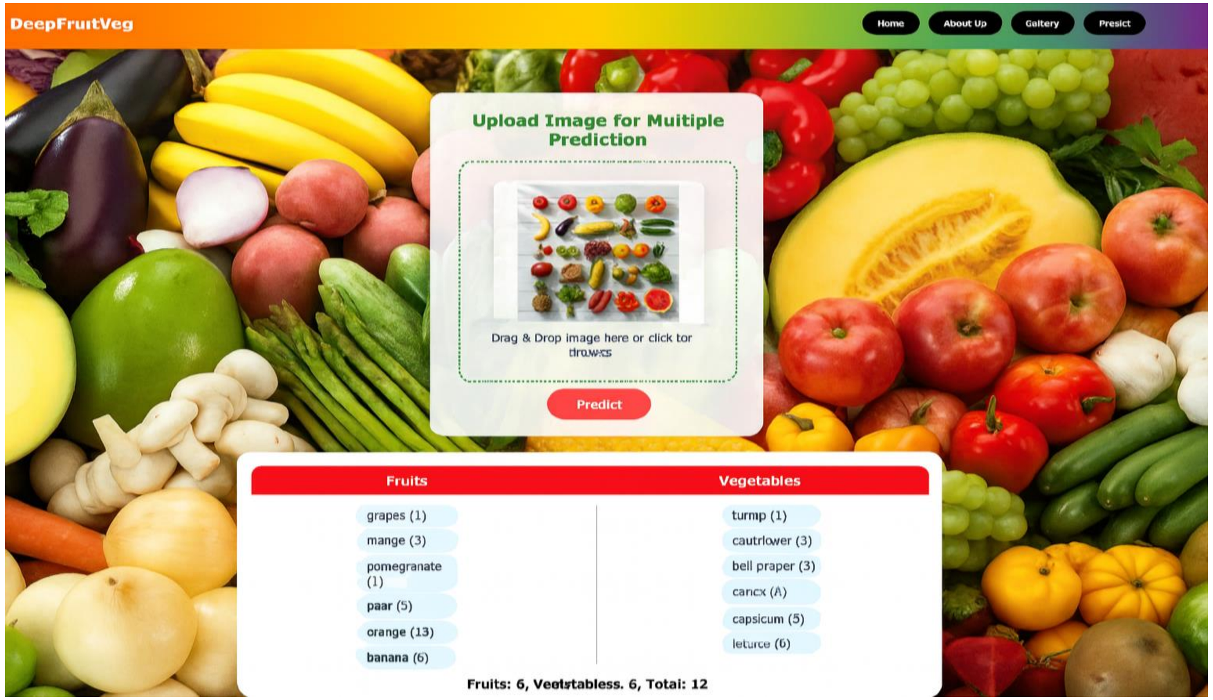
- **Training Configuration**
- Trained for **70 epochs** with batch processing to ensure stable learning across 36 classes.
- **Training Execution**
- Executed using **cnn.fit()**, which feeds image batches through the network for continuous pattern learning.
- **Optimization & Backpropagation**
- Each epoch updates model weights using forward/backward propagation, with dropout preventing overfitting.
- **Validation Monitoring**
- Validation accuracy and loss are checked every epoch to maintain balanced and reliable performance.
- **Model Saving**
- Final trained model saved via **cnn.save('fruit_model.h5')** for easy reuse and deployment.
- **Performance Summary**
- Model shows **strong, stable accuracy**, making it highly reliable for real-world fruit–vegetable identification.

Prediction & Output :

Single image prediction



Multiple image prediction

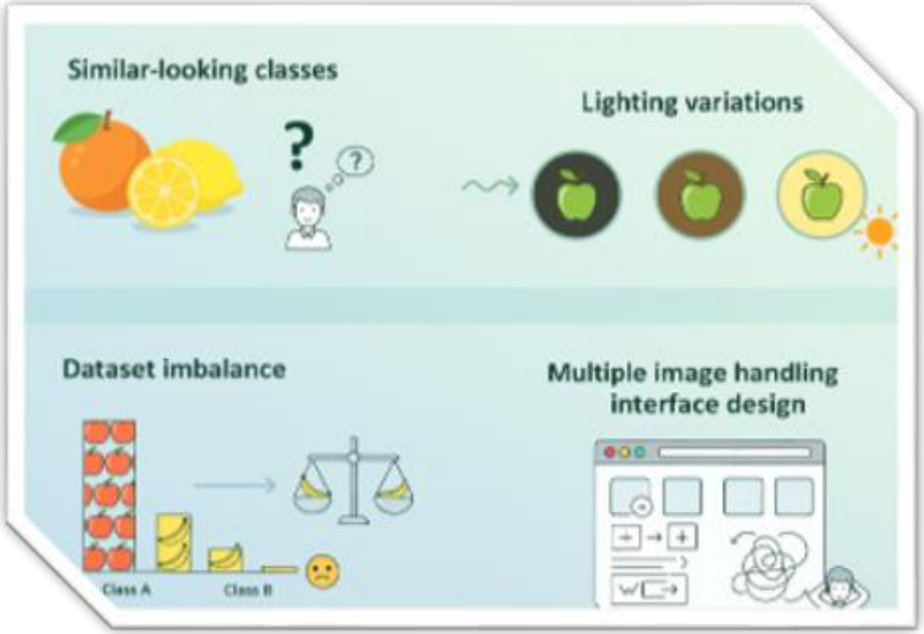


 # Calculating Accuracy of Model Achieved on Validation set
*print("Validation set Accuracy: {} %".format(training_history.history['val_accuracy'][-1]*100))*
Validation set Accuracy: 98.94061088562012 %

Here we got 98% percentage of accuracy of single image prediction and multiple image prediction

Challenges Faced

- Similar-looking classes (orange vs lemon)
- Lighting variations
- Dataset imbalance
- Multiple image handling interface design



Applications :

- Retail automation
- Smart grocery billing
- Agriculture produce sorting
- Healthy diet & nutrition apps
- Food-tech startups

Conclusion

DeepFruitVeg was created to make fruit and vegetable identification **faster, easier, and more accurate** than manual checking. Manual classification often leads to confusion, mistakes, and slow results.

What Problems It Solves

- Manual sorting is slow and tiring
- Similar-looking items create confusion
- Human errors reduce accuracy
- No quick tool for identifying food items

How Our Project Solves It

- **Smart CNN Model**

Automatically identifies fruits and vegetables with high accuracy.

- **Single & Multiple Image Support**

Handles one image or many images in a single prediction cycle.

- **Effective Preprocessing**

Cleaner, processed images improve clarity and prediction accuracy.

- **Flask Backend**

Processes images and delivers fast model predictions.

- **Simple Frontend UI**

Easy image upload and clear results for all users.

****Future Scope****

- Add more classes
- Build a mobile app
- Add nutrition value estimation
- Real-time camera capture
- Cloud deployment

THANK YOU

DEEPFRUITVEG PROJECT

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2025